

Week 1 Report

1. Why ViT?

CNNs look at small local blocks, so far-apart parts of an image connect only after many layers. ViT borrows the idea from Transformers in NLP: it cuts the image into patches and uses self-attention, so every patch can see every other patch from the first layer. Its weak point is that it has fewer built-in assumptions about images, so it needs a lot of data or a good pretrained model.

Why it suits our data

- Galaxy features like spiral arms, bars and tails are large structures spread over the image, so looking at all patches together helps.
- ViT gives one vector per patch and attention maps, which can help find parts of a galaxy without pixel-level labels.
- ViT is the backbone of DINOv2, which learns without labels. This is useful because expert galaxy labels are limited.

2. Image patchification

A Transformer needs a sequence of tokens, so the image is cut into small non-overlapping $P \times P$ squares called patches.

The image size should be a multiple of the patch size. Our DESI images are 160×160 , which is not a multiple of 14 ($160 \div 14$ is about 11.4), so the image processor resizes them to 224×224 .

3. Patch embeddings

Each flattened patch (588 numbers) goes through one shared linear layer and becomes a vector of 384 numbers, called a patch embedding or token.

- The same layer is used for every patch, so all patches are described in the same way.
- 384 is the width of the model and it stays the same through the whole network.
- At this point a token only knows its own pixels. Context from other patches comes later, in the Transformer layers.

It is like turning each word of a sentence into a word vector before a language model reads it.

4. Positional embeddings

Self-attention does not know the order of its inputs. If the patches were shuffled it would not notice, but position matters in an image, because a patch at the centre of a galaxy is different from one at the edge. So a learned position vector (384 numbers), one for each position in the grid plus one for the CLS token, is added to every token. Each token then carries both what its patch looks like and where it is.

5. CLS token

The CLS token is one extra learnable vector placed at the front of the patch sequence, so 256 patches become 257 tokens.

- It does not come from any part of the image. It starts as the same learned vector for every image.
- In each of the 12 layers it looks at all the patches through attention and collects information from them.
- At the end it is a summary of the whole image, used for classification or finding similar images.

There is only one CLS token per image, with shape (1, 384).

6. ViT Transformer architecture

ViT is an encoder-only Transformer: one block is repeated 12 times in DINOv2-small. Each block has two parts:

- **Multi-head self-attention:** every token compares itself with all other tokens and mixes in information from the ones that matter. Several heads (6 here) work in parallel, so different heads can focus on different relationships.
- **MLP:** a small two-layer network applied to each token separately.

Each part has a LayerNorm before it and a residual (skip) connection around it, which keeps training stable. The number of tokens and their size do not change, so the shape stays (1, 257, 384). The model has about 22 million parameters.

7. CLS vs patch tokens

- **CLS token:** 1 per image, shape (1, 384). A global summary, good for classification, similarity search and clustering, but it loses where things are.
- **Patch tokens:** 256 per image, shape (1, 256, 384), which form a 16×16 grid of shape (1, 16, 16, 384). Each describes a 14×14 region together with some context from the rest of the image, so they are good for locating things, but there are many vectors per image.

For galaxies, the CLS token suits questions about the whole galaxy, like "smooth or featured?" or "which galaxies look alike?". The patch tokens keep position information, so they are needed to find a spiral arm, a bar or a nearby companion.

8. What DINOv2 is

DINOv2 is a family of Vision Transformers from Meta trained with self-supervised learning, using only images and no labels. The goal was general-purpose image features that work well without extra training.

- **Student and teacher:** two copies of the network, where the teacher is a slowly updated average of the student.
- **Image level (DINO):** the student sees different crops of an image and must match the teacher, so it learns that different views of the same thing should look alike.
- **Patch level (iBOT):** some patches are hidden from the student and it must predict what the teacher sees there, which helps it learn good local features.

It comes in four sizes (small, base, large, giant), all with patch size 14. We use dinov2-small (12 layers, 6 heads, 384 dimensions, about 22 million parameters) frozen, so its weights are never changed.

9. What DINOv2 outputs

- **last_hidden_state:** (1, 257, 384), which is 1 CLS token + 256 patch tokens.
- **CLS token:** (1, 384), taken as last_hidden_state[:, 0, :].
- **Patch tokens:** (1, 256, 384), taken as last_hidden_state[:, 1:, :]. Token p is at row $p \div 16$ and column $p \bmod 16$.
- **pooler_output:** (1, 384), a copy of the CLS token.

The numbers mean nothing to a human alone, but similar images give similar vectors, so we can compare galaxies with cosine similarity.

10. How DINOv2 will be used in this project

DINOv2 is used as a frozen feature extractor on DESI DR10 galaxy images (160×160, bands g, r, i, z) with Galaxy Zoo labels:

1. **Prepare:** stretch each band, combine i, r, g into a colour image, resize to 224×224 and normalise.
2. **Extract:** get the CLS token and 256 patch tokens for each galaxy.
3. **Explore:** check if similar galaxies get similar embeddings (PCA, similarity search).
4. **Predict:** train a small model on the embeddings to predict Galaxy Zoo labels.
5. **Locate:** use patch tokens and attention maps to find galaxy parts and make masks.

What I found in Week 1

- The first 200 DESI images I streamed are mostly empty sky or stars. Only a few contain a clear galaxy, and I used sample 9, which has a galaxy with a curved tail.
- The output shapes matched what I expected: (1, 257, 384) for all tokens, (1, 384) for CLS and (1, 256, 384) for patch tokens.
- When I reduced the patch tokens of the galaxy image to 3 colours with PCA, the patches on the galaxy formed a separate colour group from the noisy background, without any labels.
- Galaxy Zoo (gz10) has 10 morphology classes. In 500 samples they were not balanced, with Round Smooth the biggest (85) and Cigar Shaped Smooth the smallest (21).